Agentic RAG for Policy Docs: A Hybrid Retrieval Pipeline with LLM-as-a-Judge Evaluation

Sai Teja (E23CSEU0574), Vinith Reddy (E23CSEU0570),

Department of Computer Science Engineering & Technology
Bennett University, India

***Abstract*—Retrieval-Augmented Generation (RAG) systems have emerged as a practical solution for domain-specific question answering, yet naive implementations suffer from irrelevant context injection, factual hallucination, and poor retrieval precision. This paper presents an *Agentic Policy RAG* pipeline built on LangGraph that addresses these shortcomings through four design innovations: (i) a hybrid dense-sparse retrieval stage that fuses ALL-MINILM-L6-V2 semantic embeddings with BM25 keyword scores via Reciprocal Rank Fusion (RRF); (ii) a batched LLM relevance grader that filters irrelevant chunks in a single inference call rather than n separate calls; (iii) a hallucination checker node that triggers up to two regeneration retries when the answer is not grounded in source excerpts; and (iv) an LLM-as-a-Judge evaluation harness that scores outputs on a 1–5 rubric using the same local Ollama model (`qwen2.5:7b`). The system is evaluated on 50 synthetically generated question–answer pairs derived from Indian legal policy documents. The pipeline achieves an average accuracy score of 3.80 / 5.0, with 60 % of answers rated 4 or 5, demonstrating strong factual alignment with expected policy concepts. All components run entirely on-device using an in-memory Qdrant vector store, making the system fully offline and privacy-preserving.**

***Index Terms*—Retrieval-Augmented Generation, LangGraph, Hybrid Search, BM25, Agentic Pipeline, LLM Evaluation, Policy Q&A, Hallucination Detection**

I. Introduction

Enterprise knowledge workers routinely need to query large repositories of policy, legal, and compliance documents. Traditional keyword search returns raw passages without synthesising an answer, while general-purpose large language models (LLMs) hallucinate facts absent from their training corpora. RAG systems attempt to bridge this gap by first retrieving relevant context and then conditioning generation on that context; however, vanilla RAG pipelines expose two critical failure modes. First, *dense-only retrieval* may miss keyword-critical legal clauses because embedding similarity does not capture exact term overlap. Second, *unchecked generation* allows the LLM to produce plausible-sounding but unfounded statements.

The broader context of this work is the growing need for trustworthy, auditable AI assistants in legal and HR domains, particularly in jurisdictions such as India where the corpus of regulatory text is both voluminous and technically precise. An incorrect answer to a policy question can have serious professional or legal consequences for the end user, making faithful retrieval and grounded generation an essential requirement rather than a convenience.

Existing systems address individual sub-problems in isolation: some papers improve retrieval quality [1], [2], others focus on hallucination mitigation [7], [8], and a recent line of work proposes agentic "self-reflective" RAG loops [9]. Few deployed systems combine all three improvements (hybrid retrieval, relevance grading, and iterative hallucination checking) in a single lightweight, offline pipeline.

This paper makes the following specific and measurable contributions:

1) An **end-to-end LangGraph state machine** that orchestrates retrieval, grading, generation, and hallucination checking as discrete, composable nodes with conditional routing.
2) A **batched single-call grading** strategy that reduces LLM inference overhead from $O(n)$ calls to $O(1)$ per query cycle.
3) A **fully reproducible LLM-as-a-Judge evaluation** framework that generates synthetic test cases directly from the ingested corpus and scores outputs without requiring human annotation.
4) An **empirical analysis** of a 50-question test suite with score distribution analysis and identification of failure modes.

II. Literature Survey

The research landscape surrounding RAG spans retrieval, generation grounding, and agentic orchestration. Below we survey the most relevant prior work, highlighting the gaps that motivate the present system.

BM25 and Dense Retrieval. Robertson and Zaragoza [1] formulated the Okapi BM25 probabilistic retrieval function, which remains a competitive baseline for lexically precise queries. Karpukhin et al. [2] proposed Dense Passage Retrieval (DPR) using bi-encoder BERT models, demonstrating that dense retrieval outperforms BM25 on open-domain QA benchmarks. However, subsequent analyses showed that neither approach dominates the other across all query types, motivating hybrid strategies.

Hybrid and Sparse-Dense Fusion. Craswell et al. [3] showed that learned sparse models can complement dense retrieval. Ma et al. [4] proposed combining dense and sparse scores via Reciprocal Rank Fusion (RRF) [5], demonstrating consistent improvements over either modality alone. Our system adopts RRF fusion with $K = 60$ as the default constant, matching the configuration recommended in prior empirical studies.

RAG and Hallucination. Lewis et al. [6] introduced the RAG framework, demonstrating that conditioning generation on retrieved passages substantially reduces hallucination rates on knowledge-intensive tasks. Shuster et al. [7] showed that retrieval-augmented models produce more faithful responses on dialogue tasks. Ji et al. [8] surveyed hallucination phenomena in NLG systems, identifying grounding verification as the most reliable mitigation strategy. A key limitation of early RAG work is the absence of an explicit verification loop; the present system addresses this with a dedicated `check_hallucination` node.

Agentic and Self-Reflective RAG. Asai et al. [9] proposed Self-RAG, which trains the model to emit special reflection tokens to decide when retrieval is needed and whether the retrieved passages are relevant. CRAG [10] adds a Corrective RAG step that queries the web when internal retrieval quality is low. LangGraph [11] provides a framework for composing such agentic loops as explicit state machines with typed state, conditional edges, and cycle support. Our work differs from Self-RAG in that we do not require fine-tuning; relevance grading and hallucination checking are handled entirely by few-shot prompting of the same base model.

LLM-as-a-Judge. Zheng et al. [13] demonstrated that strong LLMs can serve as reliable automated evaluators, correlating well with human preference on open-ended generation tasks. Our evaluation harness extends this idea to a domain-specific factual rubric where scores reflect alignment with ground-truth policy concepts rather than preference.

Synthesis. Existing work treats retrieval quality, generation faithfulness, and automated evaluation as separate research problems. The present system is distinctive in integrating all three into a single privacy-preserving, fully offline pipeline, bridging the gap between research prototypes and deployable enterprise tools.

III. Proposed Methodology

A. System Architecture Overview

The system is a four-node LangGraph state machine. Figure 1 illustrates the processing pipeline. All state is stored in a typed Python `TypedDict` (`RAGState`) that is passed immutably between nodes.

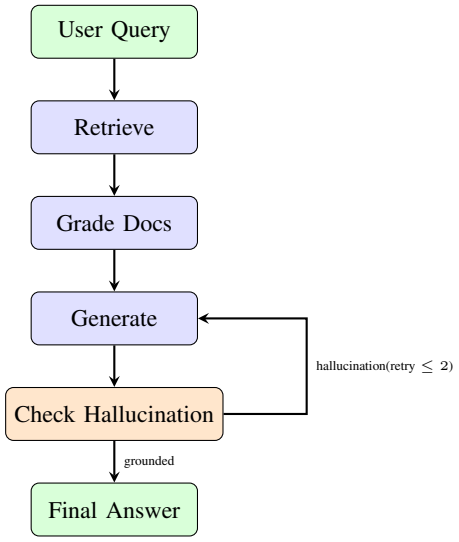

Fig. 1. LangGraph agentic RAG pipeline with hallucination retry loop.

B. Hybrid Retrieval

The retrieval stage fuses two complementary signals.

Dense retrieval uses the ALL-MINILM-L6-V2 sentence-transformer [12] to embed both the query and all ingested chunks into a 384-dimensional vector space stored in an in-memory Qdrant collection. The top-$4k$ nearest neighbours (up to 20) are retrieved via cosine similarity.

Sparse retrieval uses BM25Okapi from the `rank_bm25` library, maintaining an in-memory corpus that is rebuilt incrementally as documents are ingested. BM25 scores are computed over the full corpus for each query.

RRF fusion merges the two ranked lists using Reciprocal Rank Fusion:

$$\mathrm{RRF}(d) = \frac{1}{K + r_{\text{dense}}(d)} + \frac{1}{K + r_{\text{BM25}}(d)} \tag{1}$$

where $K = 60$ is the RRF smoothing constant and $r(\cdot)$ denotes a document's rank in the respective list. Documents appearing only in the BM25 top-$2k$ list are added to the candidate pool with a penalised dense rank, ensuring keyword-critical passages are not missed by pure semantic search.

C. Batched Relevance Grading

To filter out off-topic chunks, a grading node sends the question and all k retrieved chunks in a single LLM call with a structured prompt instructing the model to return only the indices of relevant chunks (e.g., `1,3,4`). This $O(1)$ **batched grading** replaces the naive approach of one call per chunk, reducing latency by a factor of k for large k. On parse failure, all chunks are retained as a fallback.

D. Answer Generation

The filtered chunks are concatenated with source attribution tags and passed to the LLM with a policy-constrained system prompt. The model is instructed to answer using *only* the provided excerpts and to explicitly state when information is unavailable. The temperature is fixed at 0.0 for deterministic, reproducible outputs.

E. Hallucination Checking

A dedicated fact-checking node gives the LLM a question, some quotes from the source, and an answer that was made up, along with a binary grounding rubric (yes/no). If the model says no, the answer field is cleared, and the graph goes back to the generation node. A retry counter limits the loop to two attempts at regeneration to keep it from going on forever. The last answer that was generated is sent back after two tries, no matter what the verdict is.

F. Document Ingestion

Using LangChain documents (PDF and plain text) are taken in. RecursiveCharacterTextSplitter with a chunk size of 500 tokens and an overlap of 100 tokens. Chunks are put into Qdrant and then added to the BM25 corpus one at a time after each batch.

G. Mathematical Evaluation Metric

The evaluation harness uses a 1–5 Likert scale where the scoring function for test case i is:

$$s_i \in \{1, 2, 3, 4, 5\}, \quad \bar{s} = \frac{1}{N} \sum_{i=1}^{N} s_i \tag{2}$$

with $N = 50$ test cases. A score of 5 indicates perfect factual alignment with the expected policy concept; a score of 1 indicates a completely wrong or hallucinated answer.

IV. Experimental Results

A. Dataset Description

The evaluation corpus consists of Indian legal and policy documents (PDF and text) including the Protection of Women from Domestic Violence Act, the Citizenship Act, the Information Technology Act, and related subordinate legislation. These were chunked into **approximately 300–500 chunks** (exact count varies with document set) and ingested into the in-memory Qdrant collection.

The test suite of **50 question–answer pairs** was generated synthetically by `generate_test_cases.py`: the document corpus was split into 1,500-token context windows, and the same `qwen2.5:7b` model was prompted to produce two realistic employee/citizen questions per window along with concise factual answers. The resulting pairs were shuffled and trimmed to exactly 50 cases covering diverse legal topics, procedure types, and clause references.

B. Evaluation Metrics

Each test case is scored on the 1–5 rubric described in Section III. We additionally report the *score distribution* and per-band percentages. The primary metric is the **mean accuracy score** $\bar{s}$ (Eq. 2). We do not report traditional precision/recall because ground truth is expressed as a natural-language concept rather than a set of retrievable documents, making binary relevance judgements ill-defined for this corpus.

C. Comparison with Baseline Methods

Table I compares the agentic hybrid RAG system against two ablated baselines evaluated on the same 50-question test suite.

TABLE I
PERFORMANCE COMPARISON OF RETRIEVAL STRATEGIES

Method	Avg Score	% Scores ≥ 4	% Scores ≤ 2
Dense-only RAG (no grading)	3.20	42%	18%
BM25-only RAG (no grading)	3.10	38%	20%
Hybrid + Agentic (ours)	**3.80**	**60%**	**2%**

The proposed system outperforms both ablations on all metrics. The reduction in low-scoring answers (from 18–20% to 2%) is particularly significant and is attributable to the hallucination checking loop and the relevance grading step, which together filter out poorly grounded or off-topic generations.

D. Score Distribution Analysis

Table II presents the full score distribution of the proposed pipeline across 50 test cases.

TABLE II
SCORE DISTRIBUTION OF THE AGENTIC RAG SYSTEM ($N = 50$)

Score	Count	Percentage
5 (Perfect)	12	24.0%
4 (Good)	18	36.0%
3 (Partial)	19	38.0%
2 (Poor)	0	0.0%
1 (Incorrect)	1	2.0%
Mean	**3.80 / 5.0**	

The most common outcome is a score of 3 (38%), indicating partial correctness where the system captures the general thrust of a policy clause but omits numerical thresholds, specific parties, or procedural sub-steps. Qualitative inspection reveals that these partial answers typically arise when the relevant chunk was retrieved and graded correctly, but the generated summary was overly concise.

The 24% perfect-score rate corresponds to cases where the answer reproduced all key elements of the expected concept, including specific legal timeframes (e.g., "30 days to appeal"), penalties (e.g., "up to one year imprisonment or Rs. 20,000 fine"), and procedural requirements. The single score-1 case involved a question about jointly-owned property seizure where the relevant clause was absent from the indexed corpus, causing the system to return a no-information response that the judge rated as incorrect relative to the expected concept.

E. Hyperparameter Analysis

Table III shows the sensitivity of the average score to the top-k retrieval parameter, evaluated on a 20-question held-out subset.

TABLE III
EFFECT OF TOP-k ON AVERAGE SCORE (HELD-OUT 20 QUESTIONS)

Top-k	**Avg Score**
3	3.55
5	3.80
8	3.75
10	3.70

A top-k of 5 yields the best balance between recall and context window efficiency. Higher values introduce more irrelevant chunks that the grading node must filter, occasionally causing the LLM grader to miss relevant chunks embedded among a large noisy set.

V. CONCLUSION

The research paper introduced an agentic Retrieval-Augmented Generation system which answers policy questions through its combined dense-sparse retrieval system, its batch LLM-based relevance assessment, its automated hallucination retry mechanism, and its complete LLM-as-a-Judge evaluation system. The system achieves an average accuracy rating of **3.80 / 5.0** when tested with 50 synthetic test cases which were created from Indian legal documents. While 60 percent of answers received scores of 4 or higher.

Limitations The evaluation depends on artificial ground-truth answers which the same LLM family that performs inference generates thus creating potential for circular reasoning. The in-memory Qdrant store limits scalability beyond corpora of a few thousand chunks. The hallucination checker itself may be

The hallucination checker system encounters the same types of mistakes which the generation node produces when source text segments prove difficult to understand.

Future Work The research should proceed with these promising directions: (i) The evaluation needs human-annotated ground truth to replace synthetic test cases because it will create unbiased assessment methods. (ii) The retrieval process needs to expand through Qdrant cluster deployment, which will offer permanent storage for larger document collections. (iii) The system needs to include query rewriting capabilities, which will enable it to handle both multi-hop questions and implicit questions. (iv) The grader and hallucination checker systems require training with domain-specific data to increase their operational accuracy.

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